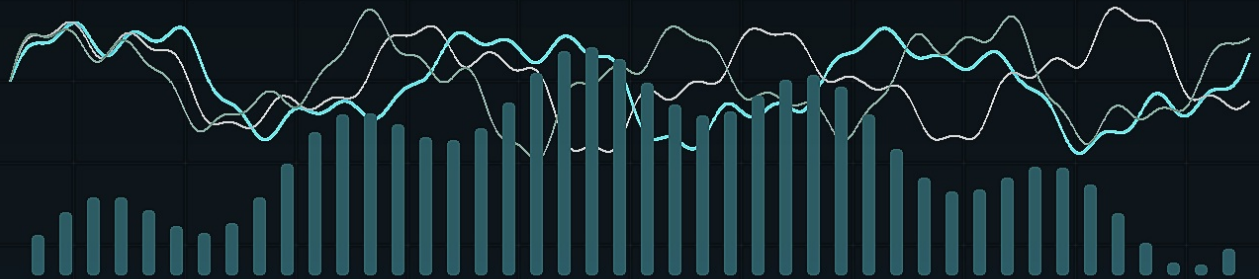


051-LAB ENGINEERING RECORDS

AXIOM-DSP

SESSION WORK LOG

JamesDSP / EEL2 Audio DSP



Current Baseline: v4.1.4.11

Stability First | Measurement Second | Experimentation Third

Repository workflow, qualification gates, Pi runbooks, and Codex orchestration record.

DOCUMENT CLASS: PUBLIC REPOSITORY LOG

NO AUDIO CAPTURES OR PRIVATE MATERIAL INCLUDED

Run 001 - AI Development Ecosystem Operating Framework

Date: 2026-06-03

Status: Completed

Scope: Documentation and workflow architecture only.

What Was Implemented

- Converted the local axiom-idea.html concept into repository-ready operating docs.
- Added the Codex, Pi, user, and external LLM role model.
- Added the system roadmap for Core, Reference, Immersive, Night, Studio Path, Labs, Knowledge, and Qualification.
- Added the ordered implementation plan for building the Axiom operating system.
- Added Labs policy, GitHub issue templates, Knowledge structure, Labs/review templates, and the system status dashboard.
- Updated the documentation index, agent instructions, contributing guide, harness documentation, and task backlog.

Why It Matters

- Axiom now has a controlled development structure around the DSP instead of loose one-off work.
- New ideas have clear intake paths before they can become DSP edits.
- Labs can explore without weakening the accepted Core baseline.
- External LLM feedback can be captured as advisory input and converted into testable work.
- A fresh Codex or Pi session can quickly find the accepted baseline, open investigations, and next work.

Validation

- git diff --check passed.
- GitHub issue-template YAML parsed successfully.
- Relative Markdown links resolved.
- Stale current-baseline scan found no current-facing .10 accepted-baseline references.
- No src/ or scripts/ files changed.

Next Recommended Work

- Review, commit, and publish the documentation and workflow foundation.
- Rerun the targeted .11 Sub Harmonics map when the JDSP route is available.
- Add concrete Knowledge or Labs examples only after the first real use case exists.

Run 002 - Session Work Log PDF Workflow

Date: 2026-06-03

Status: Completed

Scope: Documentation support tooling and generated PDF copies.

What Was Implemented

- Added docs/session-work-log.md as the editable source of truth for session work summaries.
- Added tools/generate_session_work_log_pdf.py to generate one PDF page per run section.
- Generated the repository PDF copy at docs/session-work-log.pdf.
- Generated the Windows Documents PDF copy at
C:\Users\soloa\Documents\Axiom-DSP-Session-Work-Log.pdf.

Why It Matters

- The user now has one readable document for reviewing completed work outside the chat.
- The repo keeps the source log and a PDF copy, while Windows Documents keeps an easy-access mirror.
- Future sessions can append a new ## Run section and regenerate both PDF copies with one command.

Validation

- PDF generation uses only the Python standard library.
- Both destination PDFs are written by the same generator from the same source.
- Each run section starts on its own PDF page.

Next Recommended Work

- After each completed work section, append a new run entry to this source file.
- Regenerate the PDFs before final summaries or before publishing workflow changes.

Run 003 - Profile Matrix, Listening Protocol, And Release Gates

Date: 2026-06-03

Status: Completed

Scope: System architecture and workflow documentation only.

What Was Implemented

- Added docs/profile-matrix.md to define each Axiom lane, its current status, operating authority, allowed changes, and required promotion tests.
- Added docs/listening-protocol.md to define when to listen, how to set up comparisons, what material classes to use, and how to record accept/reject decisions.
- Added docs/release-gates.md to define documentation, tooling, Labs, candidate, listening, and accepted-baseline promotion gates.
- Updated docs/README.md, AGENTS.md, CONTRIBUTING.md, the operating-system implementation plan, and the task backlog so these gates are discoverable.

Why It Matters

- Future ideas now have a clear destination before code changes start.
- Listening work is separated into setup, material selection, comparison method, observation fields, and decision rules.
- The accepted baseline has explicit promotion gates, which protects Axiom Core from broad or under-tested changes.
- Profile concepts like Reference, Immersive, Night, and Studio Path can be discussed without accidentally becoming official products.

Validation

- git diff --check passed.
- Relative Markdown links resolved.
- GitHub issue-template YAML still parsed successfully.
- Stale current-baseline scan remained clean.
- No src/ or scripts/ files changed.

Next Recommended Work

- Add Pi runbooks for repeatable terminal execution.
- Then review and publish the full workflow foundation as a docs-focused PR.

Run 004 - Pi Runbooks

Date: 2026-06-03

Status: Completed

Scope: Terminal workflow architecture only.

What Was Implemented

- Added docs/pi-runbooks.md with repeatable missions for Codex/Pi sessions.
- Covered baseline audit, candidate readiness, investigation without candidate, Labs experiments, candidate qualification, listening acceptance, advisory review triage, repository housekeeping, and publication/merge.
- Updated docs/README.md, AGENTS.md, docs/system-status.md, the operating-system implementation plan, and the task backlog so the runbooks are discoverable.

Why It Matters

- A Pi session can now be opened with a narrow mission instead of vague context.
- Each runbook defines inputs, commands, outputs, stop conditions, and decisions.
- Real-host JDSP work is protected from overlapping measurements.
- Candidate creation, Labs work, advisory feedback, and publication now have repeatable terminal paths.

Validation

- git diff --check passed.
- Relative Markdown links resolved.
- GitHub issue-template YAML still parsed successfully.
- Stale current-baseline scan remained clean.
- No src/ or scripts/ files changed.

Next Recommended Work

- Review the full documentation/workflow batch.
- Commit and publish it as a docs-focused PR after approval.

Run 005 - Review, Commit, And Publish Workflow Foundation

Date: 2026-06-03

Status: Completed

Scope: Repository publication workflow for the documentation and architecture batch.

What Was Implemented

- Reviewed the full documentation/workflow scope before publishing.
- Re-ran validation for Markdown whitespace, issue-template YAML, relative links, PDF generation, stale baseline references, and accidental DSP/script changes.
- Created branch codex/docs-axiom-operating-system.
- Committed the workflow foundation with message docs(workflow): add Axiom operating system framework.
- Pushed the branch to GitHub.
- Opened PR #10: [codex] Add Axiom operating system workflow docs.

Why It Matters

- The Axiom operating-system foundation is now reviewable in GitHub instead of existing only as local edits.
- The PR keeps this broad documentation/workflow batch separate from main until reviewed and merge-approved.
- The validation record gives future agents confidence that no accepted DSP files were touched.

Validation

- git diff --check passed.
- Issue-template YAML parsed successfully.
- Relative Markdown links resolved.
- The session PDF regenerated and both PDF copies matched by SHA-256.
- pdftotool confirmed the session PDF was valid.
- The stale .10 current-baseline scan was clean.
- git diff -- src scripts | wc -c returned 0.
- GitHub reported PR #10 as clean and mergeable.

Next Recommended Work

- Review PR #10.
- Merge only after explicit approval.
- After merge, return to the roadmap and choose the next workflow refinement or targeted .11 measurement.

Run 006 - PR Review Validation Cleanup

Date: 2026-06-03

Status: Completed

Scope: Pull request validation cleanup for the documentation/workflow batch.

What Was Implemented

- Reviewed PR #10 from the committed diff rather than only the local worktree.
- Fixed committed whitespace issues in issue templates, documentation files, and repository templates.
- Updated the session-log PDF generator so generated PDF xref entries no longer create Git whitespace warnings.
- Regenerated the repository session PDF and the Windows Documents mirror.

Why It Matters

- The PR now meets the same whitespace standard that will be enforced on merge.
- The PDF generator fix prevents this exact review failure from returning in future session-log updates.
- The cleanup keeps the workflow foundation publication-ready without changing any EEL DSP scripts.

Validation

- git diff --check main..HEAD passed after cleanup.
- Issue-template YAML parsed successfully.
- Relative Markdown links resolved.
- The session PDF regenerated and both PDF copies matched by SHA-256.
- pdftools confirmed the session PDF was valid.
- No src/ or scripts/ files changed.

Next Recommended Work

- Push the amended PR branch.
- Merge PR #10 only after explicit approval.

Run 007 - Post-Merge Roadmap Completion Reconciliation

Date: 2026-06-03

Status: Completed

Scope: Repository workflow roadmap status after PR #10 merge.

What Was Implemented

- Confirmed PR #10 merged into main as commit c498688.
- Confirmed the local main branch fast-forwarded to origin/main.
- Reconciled the completed workflow-foundation items that are now live in the repository:
 - AI development ecosystem docs;
 - Codex operating guide;
 - Pi runbooks;
 - Labs policy;
 - Knowledge Base structure;
 - profile matrix;
 - release gates;
 - listening protocol;
 - GitHub issue templates;
 - task backlog;
 - session work log PDF workflow.

Why It Matters

- The PDF now has a single page that proves which roadmap foundation items are complete.
- Future Codex, Pi, and external-review sessions can distinguish completed workflow infrastructure from larger product-profile work that is only defined.
- The repository is positioned to move from workflow setup back into measured DSP investigation without losing the architecture record.

Validation

- PR #10 is merged on GitHub.
- Local main is synced with origin/main.
- Post-merge whitespace, issue-template YAML, and PDF validity checks passed.
- No src/ or scripts/ files changed in the workflow-foundation merge.

Next Recommended Work

- Refresh docs/system-status.md and docs/task-backlog.md so they no longer describe merged workflow items as pending.
- Then continue with the targeted .11 Sub Harmonics / limiter-pressure investigation or the next roadmap housekeeping task.

Run 008 - Post-Merge Reconciliation Cleanup

Date: 2026-06-03

Status: Completed

Scope: Documentation status reconciliation after the operating-system foundation merge.

What Was Implemented

- Updated docs/axiom-operating-system-implementation-plan.md with a completion summary for all eight implementation-plan sections.
- Updated docs/task-backlog.md with explicit status values for all 17 workflow-foundation tasks.
- Updated docs/system-status.md so the operating-system foundation is marked as merged instead of drafted.
- Preserved the distinction between completed workflow infrastructure and larger product lanes that are defined but not yet built.

Why It Matters

- A fresh agent can now tell that the implementation-plan phase is functionally complete.
- The backlog no longer points at already-merged publication work as the next priority.
- The status dashboard now sends the project toward full-state assessment, targeted .11 investigation, or Knowledge/Labs examples instead of stale PR work.

Validation

- Stale "drafted" and "review/publish" language was removed from active status docs.
- Historical session-log entries were left intact because they describe earlier run states.
- git diff --check passed after the reconciliation edits.
- The session PDF regenerated successfully in the repo and Windows Documents mirror.
- No EEL DSP scripts changed.

Next Recommended Work

- Review and commit the reconciliation cleanup.
- Then assess the full Axiom state before choosing the next DSP or workflow task.

Run 009 - Session Work Log Cover Upgrade

Date: 2026-06-04

Status: Completed

Scope: Documentation presentation upgrade only.

What Was Implemented

- Added deterministic technical-manual cover artwork for the session work-log PDF.
- Added tools/generate_session_cover_art.py so the cover asset can be regenerated locally.
- Updated tools/generate_session_work_log_pdf.py to embed the cover image as page 1 when the asset exists.
- Kept searchable cover text in the PDF without adding unsupported audio-quality claims.
- Regenerated the repository PDF and Windows Documents mirror.

Why It Matters

- The session work log now reads more like an official Axiom engineering record.
- The cover is reproducible and does not depend on private images, source audio, or external design files.
- Normal PDF refreshes still work from the repo and keep both PDF copies synchronized.

Validation

- Cover artwork generated successfully.
- PDF regenerated successfully with the cover as page 1.
- Repo and Windows PDF copies matched by SHA-256.
- PDF text extraction found the cover title.
- No EEL DSP scripts changed.

Next Recommended Work

- Review the full local documentation batch.
- Then assess the current Axiom system state before starting the next major agentic-system functionality set.

Run 010 - Agentic Engineering Blueprint Started

Date: 2026-06-04

Status: Completed

Scope: New living blueprint for the future Axiom agentic engineering system.

What Was Implemented

- Added docs/axiom-agentic-engineering-blueprint.md as the source document for the agentic-system architecture.
- Added tools/generate_agentic_blueprint_pdf.py to generate a matching PDF.
- Updated docs/README.md so the blueprint is discoverable from the documentation index.
- Generated the repository PDF at docs/axiom-agentic-engineering-blueprint.pdf.
- Generated the Windows Documents mirror at C:\Users\soloa\Documents\Axiom-DSP-Agentic-Engineering-Blueprint.pdf.
- Captured the current gap: Axiom has Pi roles and commands, but not native Codex Axiom agents, subagents, or commands yet.

Why It Matters

- The agentic-system idea now has its own durable architecture document instead of living only in chat.
- The blueprint separates the Human Authority, Codex Orchestration, Pi Execution, and Knowledge/Advisory layers.
- The next major functionality set can be planned against a written target: agent registry, Codex skill pack, knowledge retrieval, role reviews, approval gates, and handoff protocol.

Validation

- Blueprint PDF generated successfully in the repo and Windows Documents mirror.
- PDF text extraction found the blueprint title.
- git diff --check passed after adding the blueprint files.
- No EEL DSP scripts changed.

Next Recommended Work

- Review the blueprint and decide the first implementation phase for the Axiom Agentic Engineering Layer.

Run 011 - Axiom Agentic Engineering Layer v1

Date: 2026-06-04

Status: Completed

Scope: Repo-tracked native Codex orchestration layer and safe helper tooling.

What Was Implemented

- Added repo-tracked Axiom Codex skill source under tools/codex-skills/axiom-engineering/.
- Added skill references for role registry, handoff protocol, approval matrix, Knowledge policy, and helper CLI usage.
- Added tools/install_axiom_codex_skill.py with --dry-run and explicit --install modes.
- Added tools/axiom-codex/axiom_codex.py with status-summary, ready-check, agent-review, and knowledge-query commands.
- Added the local-only Knowledge source-index schema and a repo-safe source-note template.
- Updated the agentic blueprint, documentation index, tool inventory, Knowledge docs, and system status.

Why It Matters

- Axiom now has a native Codex-side orchestration layer in source form without replacing the Pi execution harness.
- Codex can summarize state, frame multi-role reviews, and search Knowledge safely while real-JDSP and candidate workflows remain delegated to Pi.
- The skill can be previewed before installation, keeping local Codex changes explicit and reversible.
- Knowledge integration now supports both public repo-safe notes and a private local source index.

Validation

- Axiom Codex helper CLI smoke tests passed.
- Skill install dry run listed the expected files without modifying ~/.codex.
- Python helper scripts compiled.
- git diff --check passed.
- Markdown relative links resolved.
- Blueprint and session PDFs regenerated successfully in repo and Windows Documents mirrors.
- No EEL DSP scripts changed.

Next Recommended Work

- Review the local v1 agentic layer.
- Install the Codex skill only after explicit approval.
- Then decide whether to deepen Knowledge ingestion or start a measured Axiom DSP assessment.

Run 012 - Full Axiom System Assessment

Date: 2026-06-04

Status: Completed

Scope: Current-state assessment across Core DSP, Qualification, Agentic tooling, Knowledge, and repository housekeeping.

What Was Implemented

- Added docs/axiom-full-state-assessment-2026-06-04.md.
- Linked the assessment from docs/README.md.
- Updated docs/system-status.md with the latest assessment summary and next actions.
- Updated docs/task-backlog.md so the full-state assessment is marked as the current checkpoint instead of pending work.
- Captured a validation snapshot for the accepted v4.1.4.11 Core baseline, Pi harness, material corpus, candidate readiness, and Codex helper layer.

Why It Matters

- Axiom now has a single current-state assessment that says what is accepted, what is strong, what is still weak, and what should not happen next.
- The assessment keeps .11 as the accepted baseline while making clear that candidate readiness does not justify creating .12 without a scoped hypothesis.
- The document separates the local agentic layer's real current state from the future target: source-ready and validated, but not installed as a native Codex runtime yet.
- The roadmap can now move from broad operating-system buildout into focused follow-through: close the .11 Sub Harmonics boundary, optionally install the Codex skill, and populate Axiom Knowledge.

Validation

- scripts/validate_axiom_static.sh src/axiom_binaural_dsp_v4.1.4.11.eel passed.
- python3 -m unittest discover -s tests -p 'test_*.py' passed with 159 tests.
- npm test in tools/axiom-team passed with 22 tests.
- node tools/axiom-team/bin/axiom-team.mjs doctor passed.
- node tools/axiom-team/bin/axiom-team.mjs corpus-status --strict-metadata passed with 14 tracks and required metadata coverage.
- python3 scripts/evaluate_axiom_candidate_readiness.py reported READY.
- python3 tools/axiom-codex/axiom_codex.py ready-check passed.
- git diff --check passed.
- No EEL DSP scripts changed.

Next Recommended Work

- Review the local docs/tooling batch.
- Commit and publish the batch when explicitly approved.
- Install the Axiom Codex skill only after explicit approval.
- Run the targeted .11 Sub Harmonics map when the JDSP route is available.
- Add the first lawful Axiom Knowledge source notes.

Run 013 - Local Docs And Agentic Tooling Pre-Commit Audit

Date: 2026-06-04

Status: Completed

Scope: Pre-commit review cleanup for the local documentation and agentic-tooling batch.

What Was Implemented

- Audited the current local diff and untracked files before any commit or publication.
- Confirmed the active diff remains documentation and tooling only.
- Updated the Axiom Codex helper agent-review scaffold so it no longer emits weak TBD placeholders.
- Updated docs/system-status.md to describe the agentic blueprint, skill source, and helper CLI as source-ready instead of merely drafted.
- Removed generated Python cache directories created by validation runs.

Why It Matters

- The local batch is closer to review-ready without changing the accepted DSP baseline.
- The agent-review helper now makes clear that role findings must be completed before the scaffold can be used as evidence.
- The system-status wording better distinguishes the real current state from the future target: source-ready local tooling exists, but the Codex skill is not installed yet.
- The repo tree is cleaner for review because generated runtime cache files were removed.

Validation

- Axiom Codex skill validation passed with quick_validate.py.
- python3 tools/axiom-codex/axiom_codex.py agent-review --topic "local docs tooling pre-commit audit" produced the updated scaffold.
- git diff --check passed.
- Python helper scripts compiled.
- python3 tools/axiom-codex/axiom_codex.py ready-check passed.
- No EEL DSP scripts changed.

Next Recommended Work

- Review the local docs/tooling batch as the candidate commit scope.
- Commit and publish only after explicit user approval.
- Keep DSP work paused until the documentation/tooling batch is safely recorded or the user explicitly redirects.

Run 014 - Agentic Foundation Commit And Draft PR

Date: 2026-06-04

Status: Completed

Scope: Local commit and GitHub publication for the docs/tooling agentic-foundation batch.

What Was Implemented

- Created branch codex/agentic-engineering-foundation.
- Staged only the intended documentation, generated PDF, generated cover-art, Knowledge template/schema, Axiom Codex helper, Codex skill source, and PDF-generator tooling files.
- Committed the batch as cef0fee with message Add agentic engineering foundation.
- Pushed the branch to origin/codex/agentic-engineering-foundation.
- Opened draft PR #11: <https://github.com/051-lab/axiom-binaural-dsp/pull/11>.

Why It Matters

- The agentic engineering foundation is now available on GitHub for review from other devices and sessions.
- Publication happened through a branch and draft PR instead of direct main mutation.
- Merge remains a separate explicit approval gate.
- The accepted Axiom Core DSP baseline remains unchanged.

Validation

- gh auth status confirmed GitHub authentication for 051-lab.
- git diff --cached --check passed.
- git diff --check passed.
- Python helper scripts compiled.
- Axiom Codex skill validation passed.
- Markdown relative-link validation passed.
- npm test in tools/axiom-team passed with 22 tests.
- python3 -m unittest discover -s tests -p 'test_*.py' passed with 159 tests.
- git diff -- src scripts | wc -c returned 0.
- PR #11 is open as a draft and GitHub reports it as mergeable.

Next Recommended Work

- Review PR #11.
- Merge only after explicit user approval.
- After merge, install the local Axiom Codex skill only if the user approves that runtime change.
- Then resume targeted .11 measurement or Axiom Knowledge population.